# ORVILLE ROCKWELL Jr.

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## EDUCATION

**Master's in Applied Artificial Intelligence** | Northeastern University *Expected 2027*
*Research Interests: Computer vision, generative modeling, and machine learning for computational biology and autonomous systems — 3D perception, sensor fusion, diffusion-based molecular design, and closed-loop discovery.*

**Bachelor of Science – Information Technology** | Purdue University Global *March 2026*
*Chancellor's List 2026 · Asst. Dean's Honors Student Advisory Council · NSLS Honor Society*

## RESEARCH EXPERIENCE

**Research Contributor – GPS-Denied Visual Navigation** | Brown University, Prof. Benjamin Kimia *2026 – Pres.*
- Contributing to a research team developing GPS-denied drone localization, combining visual-inertial sensor fusion with RF-beacon triangulation to maintain positioning accuracy without satellite navigation.
- Own the multi-camera sensing layer: hardware bring-up of a synchronized camera array (Raspberry Pi 5, Arducam Camarray HAT), camera calibration and frame-level time synchronization, with calibration routines and tests contributed to the group's repository.

**Independent Research – Calibration-Aware Active Learning for Closed-Loop Discovery** *Preliminary result*
- Simulated a closed-loop ("self-driving lab") discovery cycle over 2,311 PTP1B inhibitors from ChEMBL to test whether calibrating the scoring model reduces the experiments needed.
- Negative result over 8 seeds (35.5 ± 3.5 vs. 35.3 ± 4.2 hits at a 360-experiment budget); traced to resolution loss: isotonic regression collapses 1,844 distinct scores into 74, making batch selection arbitrary among ties.

**Independent Research – Classifier-Guided Diffusion for Post-Stroke Drug Design** | Northeastern University *In progress*
- Designing a diffusion-based molecular generator whose denoising steps are steered by property-classifier gradients for blood-brain-barrier penetration, target activity and toxicity, aimed at post-stroke glial scar repair.
- Guidance classifiers trained on noised inputs plus a separate held-out judge keep the generator from exploiting its own guide; all scorers to be evaluated on scaffold splits.
- Originated as the individual research project for AAI 6600 (Applied AI), Prof. Elena Izotova, at Northeastern University.

## MANUSCRIPTS IN PREPARATION

**Rockwell Jr., O.** When calibration breaks the loop: Score resolution and batch selection in closed-loop molecular discovery. *Manuscript in preparation; targeting a 2027 machine learning workshop.*

## WORK EXPERIENCE

**Software Onboard Engineer** | iTribe (B2C Blockchain & Fintech) *2023 – 2026*
- Built and shipped full-stack features across product rollouts spanning the Middle East, Africa, and North America.
- Led QA smoke and regression testing, sustaining a 90%+ release-readiness rate.

**Software Engineering** | RICI; MySudra *Earlier*

## PROJECTS

**Does Human Alignment Make a Network's Confidence More Reliable?** | Independent project, Northeastern University *Nov. 2026*
- Tested on 1,500 ClickMe images and eight pretrained ImageNet networks, with hypotheses written down before any model was run: the accuracy–human-alignment trade-off reproduced (Spearman −0.74), but image-level alignment added no information about errors beyond confidence in any model.

**Deep Perception & Sensor Fusion, NUWave ROV** | Northeastern Marine Robotics (team) *2026 – Pres.*
- Developing deep-learning object detection and visual–IMU sensor fusion for the ROV's ROS perception stack, including the recognition node and its automated tests, for the MATE ROV competition.

**Team Lead – Volatility-Aware Distributional Forecasting (VAD-LSTM)** | AAI 6640, Prof. Behnaz Merikhi *2026 – Pres.*
- Leading a four-person team building a recurrent network that forecasts a full probability distribution, extending the GARCH–LSTM equivalence of Zhao et al. (2024) with an in-cell variance recursion and a Student-t output head; own the ten-year dataset and evaluation (purged walk-forward splits, isotonic recalibration, conformal coverage).

**Leakage-Safe Evaluation of Learned Signals** | AAI 6610, Prof. Zhe (Richard) He *2026*
- Evaluation study on noisy time series using purged, embargoed cross-validation and multiple-testing correction, with every trial logged.

**CAD-LLM Platform · APML — Analytical Predictive ML Platform**
- CAD-LLM: LLM-backed REST pipeline converting natural-language specifications into structured CAD parameters. APML: end-to-end predictive pipeline benchmarking TensorFlow architectures on accuracy and inference latency.

## TECHNICAL SKILLS

**Languages:** Python, C++, Rust, SQL, JavaScript, HTML, R, Ruby
**Methods:** Active learning, calibration, conformal prediction, diffusion/classifier guidance, sensor fusion
**ML/AI:** PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, OpenCV, ROS, RDKit, Isaac Gym, MuJoCo, Stable-Baselines3
**Tools/Infra:** Docker, AWS, Linux, Git, REST APIs, Jira, Raspberry Pi, Wireshark